Election Integrity Dashboard: Predictive Risk Analysis

Executive Summary

This project analysed polling unit election data and election observer incident reports from the 2019 and 2023 election cycles to identify areas at risk of electoral irregularities in future elections. The datasets were cleaned, standardized, and merged using State, LGA, and ElectionYear. Inconsistencies in LGA naming conventions were resolved to ensure accurate matching across both datasets. Missing incident information was handled appropriately, and additional analytical variables were created, including Turnout Percentage, Anomaly Flag, Incident Count, Average Severity Score, Anomaly Rate, Turnout Anomaly, and Risk Score.

To estimate election risk at the LGA level, a Risk Score model ranging from 0 to 100 was developed using the following formula:

Risk Score = (Anomaly Rate × 0.4) + (Average Severity Score Scaled × 0.3) + (Incident Count Scaled × 0.2) + (Turnout Anomaly × 10)

This weighting scheme assigns the greatest importance to anomaly rates because they directly reflect unusual voting patterns at polling units. Average Severity Score and Incident Count provide evidence of the frequency and seriousness of election-related incidents, while Turnout Anomaly captures unusual voter participation patterns relative to state averages.

The analysis identified Nassarawa, Port-Harcourt, and Surulere as the three highest-risk LGAs. These LGAs emerged as high-risk areas due to their combination of elevated anomaly rates, high incident frequencies, and above-average incident severity scores. These factors consistently contributed to higher overall Risk Scores compared to other LGAs and indicate a greater likelihood of electoral irregularities if adequate monitoring measures are not implemented.

A comparison of 2019 and 2023 results revealed notable changes in electoral risk across LGAs. The comparative analysis showed that several LGAs experienced increased Risk Scores between 2019 and 2023, indicating deteriorating election integrity conditions. Conversely, Port-Harcourt emerged as the most improved LGA, followed by Surulere and Nassarawa. These LGAs recorded lower Risk Scores in 2023 compared to 2019, indicating improvements in election integrity indicators such as reduced anomalies, lower incident severity, or fewer reported incidents. While these LGAs remain among the highest-risk locations overall, their declining Risk Scores suggest measurable improvements relative to previous election cycles.

The dashboard developed for this project provides an interactive view of electoral risk across LGAs and supports decision-making through risk mapping, trend analysis, and anomaly monitoring. Key visualizations include a risk map, a 2019 versus 2023 comparison chart, an anomaly-versus-incident scatter plot, and summary indicators highlighting high-risk and improved LGAs.

No LGAs were found with zero incident reports after merging the datasets. Therefore, potential under-reporting based on the absence of incident reports could not be directly assessed using the available data. Future analyses may benefit from more granular incident reporting and expanded observer coverage to improve visibility into electoral events at the polling-unit level.

Based on the findings, it is recommended that INEC prioritize the deployment of election observers, security personnel, and monitoring resources to LGAs identified as high risk by the

model. Targeted monitoring in these areas may help reduce electoral malpractice and improve transparency during future elections.

One limitation of this analysis is that the Risk Score relies on reported incidents and observed anomalies within the available datasets. Unreported incidents or data quality issues may affect the accuracy of the risk estimates. Therefore, the results should be interpreted as indicators of potential electoral risk rather than definitive evidence of election fraud.

Overall, the Election Integrity Dashboard demonstrates how historical election data and incident reports can be combined to proactively identify high-risk areas and support data-driven election monitoring strategies.